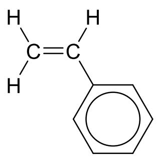
Addition Polymers

➢ Properties of addition polymer

- monomers are connected by C-C single bonds to form the polymer chain
- addition polymers are formed through ________________________________;
  in this reaction, C=C π bonds in monomers are broken, and new C−C σ bonds are formed
- addition polymer are ______________________________ because they are chemically inert;
  they cannot be broken down through hydrolysis with aqueous acid or alkali

➢ Draw a repeating unit of the addition polymer formed by the following monomer

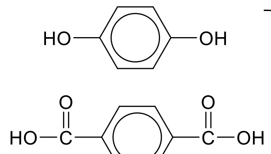


Condensation Polymers

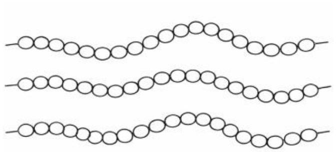
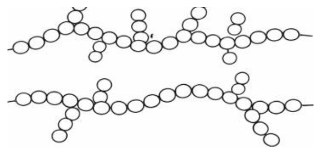
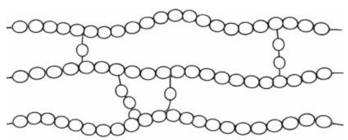
➢ Properties of condensation polymer

- monomers are connected by ester or amide bonds to form the polymer chain
- condensation polymers are formed through ____________________________________;
  in this reaction, new ester bonds or amide bonds are formed between monomers
- condensation polymer are ____________________ because they contain amide and ester bonds;
  they can be broken down through hydrolysis with aqueous acid or alkali

➢ Draw a repeating unit of the condensation polymer formed using the following monomers



Polymer structure

	Linear	**Branched**	**Cross-linked**
diagram			
interaction between polymer chains	__________________ intermolecular forces of attraction between chains because polymer chains can pack closely	___________________ intermolecular forces of attraction between chains because branching hinders polymer chains from packing closely	strong bonds between chains, e.g. covalent bonds
properties	strong, rigid	soft, flexible	hard, rigid, brittle

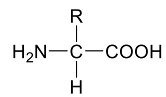
Note:

- strong vs weak: a strong polymer can withstand a large force without breaking
- flexible vs rigid: a flexible polymer can bend easily without breaking
- hard vs soft: a hard polymer can resist indentation and compression

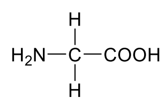
Proteins

➢ Properties of proteins

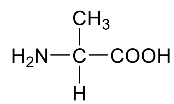
- proteins are biopolymers which contain ________________ bonds
- proteins are condensation polymers made from monomers called

_______________________ 

➢ Draw the structure of the dipeptide: Ala-Gly



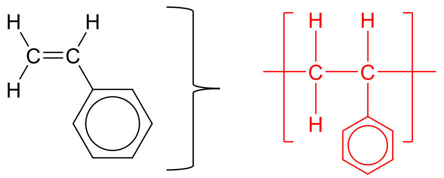
alanine (Ala)



glycine (Gly)

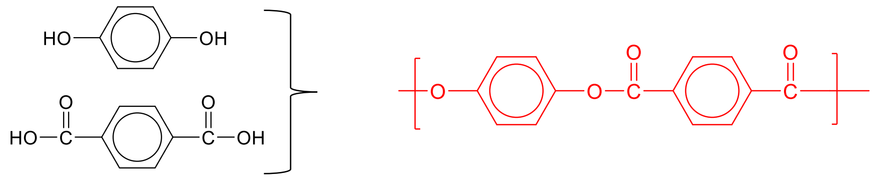
Addition Polymers

- Properties of addition polymer
  - monomers are connected by C-C single bonds to form the polymer chain
  - addition polymers are formed through ____addition reaction____;
    in this reaction, C=C π bonds in monomers are broken, and new C−C σ bonds are formed
  - addition polymer are ____non-biodegradeable____ because they are chemically inert;
    they cannot be broken down through hydrolysis with aqueous acid or alkali
- Draw a repeating unit of the addition polymer formed by the following monomer

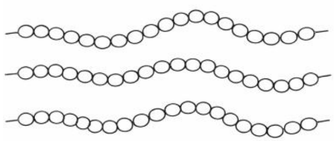
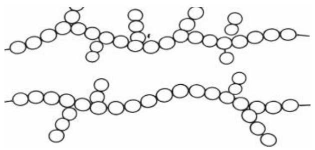
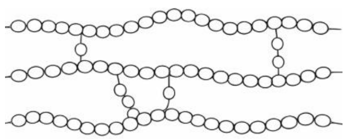


Condensation Polymers

- Properties of condensation polymer
  - monomers are connected by ester or amide bonds to form the polymer chain
  - condensation polymers are formed through ____condensation reaction____;
    in this reaction, new ester bonds or amide bonds are formed between monomers
  - condensation polymer are ____biodegradeable____ because they contain amide and ester bonds;
    they can be broken down through hydrolysis with aqueous acid or alkali
- Draw a repeating unit of the condensation polymer formed using the following monomers



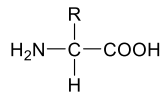
Polymer structure

	Linear	Branched	Cross-linked
diagram			
interaction between polymer chains	____stronger____ intermolecular forces of attraction between chains because polymer chains can pack closely	____weaker____ intermolecular forces of attraction between chains because branching hinders polymer chains from packing closely	strong bonds between chains, e.g. covalent bonds
properties	strong, rigid	soft, flexible	hard, rigid, brittle

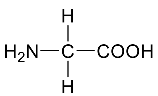
Note:
- strong vs weak: a strong polymer can withstand a large force without breaking
- flexible vs rigid: a flexible polymer can bend easily without breaking
- hard vs soft: a hard polymer can resist indentation and compression

Proteins

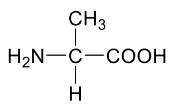
- Properties of proteins
  - proteins are biopolymers which contain ____peptide (amide)____ bonds
  - proteins are condensation polymers made from monomers called

____α-amino acids____ 

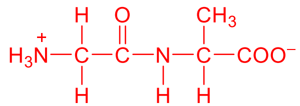
- Draw the structure of the dipeptide: Ala-Gly



alanine (Ala)



glycine (Gly)



Note: N terminus to C terminus